

CCS Operator Training

Lesson 2.1: Build and Validate a Route

Trainee guide and instructor key · September 23, 2026 · Nathan B. Smith

In 2.5 hours, operators learn to build a route in CCS Mission Planning, check it for conflicts, fix what fails, and save a plan that is ready to fly. **Representative sample:** the CCS screens and labels shown are illustrative, not the actual interface.

Lesson overview

Item	Detail
Course	CCS Operator Basic, Unit 2: Mission Planning
Lesson	2.1 Build and Validate a Route
Audience	Unmanned systems operators, all platforms
Duration	2.5 hours (0.75 classroom, 1.5 lab, 0.25 assessment)
Prerequisites	Lesson 1.2 Start a Session; Lesson 1.3 The Common User Interface
Location	CCS training lab, one station per two trainees
Training aids	CCS training build in simulation mode; scenario file <code>LP21-harbor.plan</code> ; Job Sheet 2.1

References

Reference	Purpose in this lesson
CCS Software User Manual, Section 5.3.1 Mission Planning	Authoritative procedure the lesson teaches
How-To Guide: Build a Route	The job aid trainees keep at the station after training
NAVEDTRA 130 series, Navy curriculum development	Lesson plan structure and objective format
DoD Instruction 5200.48, CUI	Marking rules if the lesson is used with real scenario data

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Learning objectives

One terminal objective, supported by four enabling objectives; each states the condition, the behavior, and the standard.

Terminal objective 2.1. Given a CCS station in simulation mode and a mission tasking, **build, validate, and save a route** that passes validation with zero conflicts, within 20 minutes, with no instructor assistance.

ID	Condition	Behavior	Standard
2.1.1	Given the Mission Planning screen	Identify the route editor, waypoint list, and validation panel	All three identified correctly
2.1.2	Given a tasking with 5 waypoints	Create a route and place waypoints in order	Waypoints within 0.1 nm of tasked positions
2.1.3	Given a route with legs defined	Set altitude or depth and speed for each leg	Values within the platform limits in the SUM
2.1.4	Given a route that fails validation	Interpret each conflict and correct the route	Route passes validation after correction

Lesson timeline

150 minutes total; more than half is hands-on.

Segment	Minutes	Method	Objectives
Introduction and motivation	10	Discussion	—
Key concepts	25	Lecture with station display	2.1.1
Instructor demonstration	10	Demonstration	2.1.2-2.1.4
Job Sheet 2.1, Parts A-C	70	Practical exercise in pairs	2.1.2-2.1.4
Break	10	—	—
Knowledge check	10	Written, closed book	2.1.1-2.1.4
Performance assessment	15	Individual, timed	Terminal 2.1

Lesson content

A route is the vehicle's travel plan: where to go, in what order, how high or deep, and how fast. CCS checks the plan before the vehicle ever moves, so mistakes are caught on the screen, not in the air or the water.

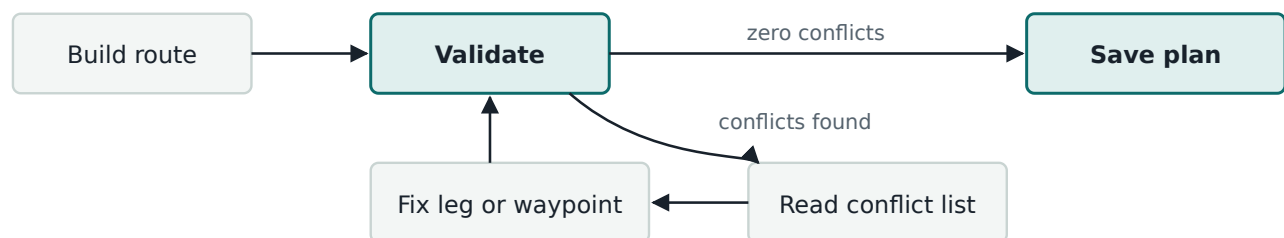
Why this matters (introduction)

Ask the class: *what happens if a driver's GPS route goes through a closed road?* The driver notices and turns. An unmanned vehicle may not. Validation is how CCS "notifies" for it, before launch.

Key terms

Term	What it actually is
Waypoint	A point the vehicle must pass through, with a position and an order number
Leg	The straight path between two waypoints, with its own altitude or depth and speed
Route	The ordered list of waypoints and legs, saved inside a mission plan
Constraint	A rule the route must obey: restricted airspace, minimum altitude, fuel or battery reserve
Validation	CCS's automatic check of the route against every constraint
Conflict	A place where the route breaks a constraint; shown in red on the map and listed in the validation panel

How validation works



The loop repeats until validation shows zero conflicts; only then is the plan saved as ready.

Common conflicts and fixes

Conflict shown	What it means	Usual fix
Airspace violation	A leg crosses restricted airspace	Move or add a waypoint to route around it
Below minimum altitude	A leg is set lower than terrain or policy allows	Raise the leg altitude
Speed out of limits	Speed exceeds what the platform can fly	Lower the leg speed to within limits
Insufficient reserve	The route is too long for fuel or battery	Shorten the route or remove a waypoint

Teaching note: platform limits differ. Point trainees to the platform supplement in the SUM (Appendix A) for their vehicle's numbers, rather than memorizing values in class.

Job Sheet 2.1: Build and validate a route

Working in pairs, trainees build the harbor-patrol route from the scenario tasking, deliberately trigger a conflict, fix it, and save. One trainee drives the station while the other reads the steps aloud; switch roles for Part C.

Tasking (scenario file LP21-harbor.plan): patrol five points around the harbor entrance, in order, returning to the start. A restricted zone sits between waypoints 3 and 4.

Safety and security: simulation mode only. Do not connect the training station to a live datalink. Scenario data is unclassified training data.

Part A — Create the route (20 min)

1. In Mission Planning, open LP21-harbor.plan.
The harbor map loads with the tasking overlay.
2. Select **Route > New**.
The route editor opens; the waypoint list is empty.
3. Click the map at each tasked point, in order, to place waypoints 1 through 5.
Each waypoint appears numbered on the map and in the list.
4. Right-click waypoint 5 and select **Close route to start**.
A leg connects waypoint 5 back to waypoint 1.

Part B — Set legs and validate (25 min)

1. Select leg 1-2. Enter the tasked altitude or depth and speed.
2. Repeat for each leg.
3. Select **Validate**.
Expected: one conflict — leg 3-4 crosses the restricted zone, shown in red.
4. Read the conflict entry in the validation panel. Write down its type and the leg it names.

Part C — Fix and save (25 min)

1. Drag a new waypoint onto leg 3-4 and move it outside the restricted zone.
The leg splits in two and turns from red to gray.
2. Set altitude or depth and speed on both new legs.
3. Select **Validate**.
Validation panel shows 0 conflicts.
4. Select **Save**. Name the plan LP21-<your initials>.

Check your work: the plan shows in the mission tree with a green "validated" status.

Knowledge check

Five questions, closed book, 10 minutes; passing is 4 of 5.

1. What is the difference between a waypoint and a leg?
2. Validation shows a leg in red with "Below minimum altitude." What do you change?
3. Why must a route pass validation before the plan is saved as ready?
4. A route fails with "Insufficient reserve." Name two ways to fix it.
5. Where do you find the speed and altitude limits for your platform?

Assessment and instructor key

Trainees pass the lesson by passing the knowledge check and meeting every critical item on the performance checklist within 20 minutes, unassisted.

Performance checklist (Terminal objective 2.1)

Item	Critical	Sat / Unsat
Opens the correct scenario file	No	
Places all waypoints in tasked order	Yes	
Sets altitude or depth and speed on every leg, within limits	Yes	
Runs validation before saving	Yes	
Identifies and corrects each conflict	Yes	
Saves plan with zero conflicts and correct name	Yes	
Completes within 20 minutes, no assistance	Yes	

Any Unsat on a critical item is a fail; remediate with Job Sheet 2.1 Part C and reassess.

Knowledge check answer key

1. A waypoint is a point to pass through; a leg is the path between two waypoints, with its own altitude or depth and speed.
2. Raise that leg's altitude until it is above the minimum, then validate again.
3. So conflicts are caught on the screen before the vehicle moves; an unvalidated plan may break airspace, terrain, or reserve limits.
4. Any two: shorten the route, remove a waypoint, or reduce distance between points (speed changes may also help, per platform).
5. The platform supplement in the SUM (Appendix A), or the How-To Guide for that platform.

Instructor notes

- Pair a stronger trainee with a newer one for Parts A and B; split pairs for the individual assessment.
- If a pair finishes early, have them add a sixth waypoint that creates a reserve conflict and fix it.
- Log recurring trainee errors in Jira against the How-To Guide; they point to steps that need clearer wording.